

SPATIAL PROPERTIES OF THE POPULATION OF MECHANORECEPTIVE UNITS IN THE GLABROUS SKIN OF THE HUMAN HAND

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SUMMARY

Impulses were recorded in low threshold mechanoreceptive afferents innervating the glabrous skin area of the human hand by means of needle electrodes percutaneously inserted in the median nerve of waking human subjects. In accordance with earlier investigations, the units were separated into 4 groups, mainly on the basis of their adaptation and receptive field properties: RA, PC, SA I and SA II units.

The extent of the receptive fields of 255 units were mapped with von Frey hairs. The fields of the RA and SA I units were of similar size with medians of 12.6 sq. mm and 11.0 sq. mm. In contrast, the fields of the SA II and PC units were about 5 and 10 times larger, respectively. A difference in size between fields from different skin regions was found only for the RA fields which were smaller the more distal the skin region. The distributions of the field sizes were positively skewed for all 4 unit types.

A model was constructed of the population of RA and SA I fields within the various regions of the hand. The model was based on (i) an estimate of the densities of units reported in a previous study¹⁴, and (ii) the sizes of the receptive fields.

The number and the spatial distribution of the fields which would be stimulated by indentations of simple and defined geometries were deduced from the model. The findings indicate that the RA and SA I units provide the information required for the psychophysical capacity of spatial analysis and discrimination in the glabrous skin of the hand.

INTRODUCTION

It can be deduced from the high density of mechanoreceptive units in the skin that most tactile stimuli engage of large number of parallel afferents, transmitting the

description of mechanical events at the body surface to the central nervous system. Temporal and spatial features of any distinctive stimulus is represented by a unique spatio-temporal pattern of impulses. It seems that a knowledge of overall afferent activity from the population of mechanoreceptive units is required for an understanding of the basic mechanisms involved in tactile sensibility. At present there is probably only one approach to this kind of knowledge, namely to construct a theoretical model of the population properties on the basis of experimental data derived from studies of single units. Only a few quantitative studies of population response have previously been carried out in the cat (e.g. refs. 2, 5, 6, 8 and 17), and one in the primate hand¹⁶.

The present study includes a theoretical synthesis of some population properties of mechanosensitive units in the glabrous skin of the human hand. It has been shown that, in all, there are 4 types of mechanosensitive units in this skin area, two rapidly adapting, RA and PC units, and two slowly adapting, SA I and SA II units. The RA and SA I units have cutaneous fields which are small and well defined. The PC and SA II units, on the other hand, have larger fields with obscure boundaries and they are sensitive to remote stimuli^{12,18-20}. The finding that the RA and SA I units are present in higher densities the higher the tactual acuity of the glabrous skin region as well as the properties of their receptive fields suggest that these two unit types account for the high tactual acuity in this skin region^{14,29}. In the present study the population of RA and SA I units in the glabrous skin was reconstructed with the purpose of elucidating the capacity of spatial discrimination of the system. The reconstruction includes a model of the overlap of the receptive fields. Moreover, the number and the spatial distribution of fields stimulated by indentations of simple and defined geometries were deduced.

The reconstruction is based on experimental data which have been published in previous reports as will be detailed below, and data on the sizes of the receptive fields given in the present report. Some of these results have been preliminarily reported^{13,29}.

METHODS

Nerve impulses of single units were recorded from the median nerve with tungsten needle electrodes²⁸. Mechanosensitive afferents with receptive fields in the glabrous skin area of the hand were randomly isolated. Fifty-two experiments were performed on 40 healthy adults, 24 on females and 28 on males. All but two subjects were 18-28 years old, the exceptions being 16 and 42 years old. Other aspects of the same sample of units have been presented in an earlier report¹⁴.

Classification of afferent units

The units were classified largely on the basis of their adaptation and receptive field properties in accordance with earlier studies^{12,18,19,29}. A rapidly adapting unit was classified as a RA unit if it had a small and well defined receptive field whereas a rapidly adapting unit with a large receptive field, poorly defined field boundaries and pronounced responses to remote taps was classified as a PC unit. Slowly adapting units

with similar field properties as the RA units were classified as SA I units whereas units with large receptive fields and a pronounced sensitivity to remote stretching of the skin were classified as SA II. The classification was often further supported by indications of several sensitivity maxima within the fields of RA and SA I units¹² as well as observations on the regularity of the sustained discharge of slowly adapting units which is high for the SA II units and low for the SA I units¹¹.

Determination of receptive field size

Mechanical stimuli were delivered with a set of nine von Frey hairs calibrated with a balance to provide the following forces (mN) when held steadily: 0.1, 0.5, 1, 2, 5, 10, 20, 50, and 100. The diameters of the threads were (mm) 0.06, 0.09, 0.19, 0.21, 0.27, 0.21, 0.21, 0.30, and 0.30, respectively. When a unit was isolated its threshold, defined as the force of the weakest von Frey hair which excited the unit was determined. Then the receptive field was assessed with a hair which gave a force of 4–5 times the threshold. In these tests the hair was prodded fairly rapidly towards various points within the receptive field. When an impulse appeared in several tests the stimulus was accepted as threshold or suprathreshold. The borders of the area from which impulses could be evoked were marked on the skin with a sharp ink pen and the field size was calculated. A minority of SA II units (about 20%) which exhibited a spontaneous discharge in all positions of the hand were discarded, since their absolute thresholds could not be measured.

It has been shown in a separate study that the fields of the RA and SA I units, as determined by this method, agreed rather well with the high sensitivity area of the receptive field and enclosed the zones of maximal sensitivity with a small clearance. These zones probably reflect the location of the end-organs¹². Hence, it seems that the present method of mapping the field, despite its simplicity, defines the (RA and SA I) fields with a sufficient precision and uniformity to serve as a basis for quantitative estimates of population properties.

Glabrous skin regions

Three separate regions of the glabrous skin area were considered: (i) the skin of the terminal phalanx distal to the vortex, referred to as the finger tip, (ii) the remaining part of the finger and (iii) the palm. This division has been considered in detail in an earlier report¹⁴.

Statistics

Non-parametric statistics were used to describe the characteristics of the unit populations (e.g. ref. 26). The statistical significance of differences between two populations was assessed using the Mann–Whitney U-test. When three populations were considered the Kruskal–Wallis one-way analysis of variance was used. The level of probability selected as significant was a value of $P < 0.1$ (two-tailed test).

Calculations were made with the aid of a programmable desk calculator (Hewlett Packard 9810 A) equipped with a plotter (9862 A).

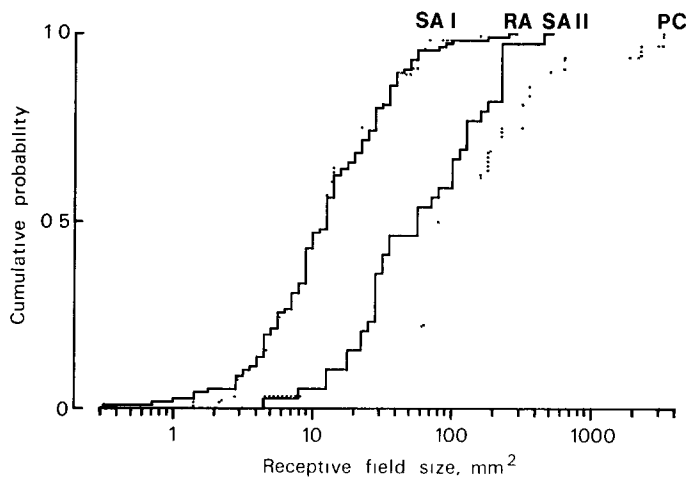


Fig. 1. Distributions of sizes of receptive fields shown as separate cumulative histograms for the 4 types of mechanosensitive units. Interrupted curves refer to the SA I and the PC units. The number of units underlying the distributions is 118 for the RA units, 66 for the SA I units, 39 for the SA II units and 32 for the PC units. Note that the abscissa is logarithmic.

RESULTS

Receptive fields

Altogether, 255 receptive fields were mapped. The shapes of the fields were essentially similar to those described in previous reports, i.e. roughly round or oval, and they were accordingly classified in two groups^{12,29}. About three-quarters of the RA and the SA I fields were oval. Two-thirds of the oval fields had their long axes roughly parallel to the longitudinal direction of the hand. The others were transversely or intermediately oriented. For all but one-fifth of the RA and the SA I units, the field, defined as described in Methods, was a single area. The exceptional fields consisted of two to four areas which were relatively closely spaced, never more than a few millimetres apart. There were no statistically significant differences between these two sets of units with regard to field size.

Receptive field size. The sizes of the receptive fields are presented in Fig. 1 which shows the cumulative distributions with a logarithmic abscissa. The RA ($n = 118$) and SA I ($n = 66$) units had fields of similar size, whereas the PC ($n = 32$) and SA II ($n = 39$) fields were considerably larger. The medians were 12.6 sq. mm, 11.0 sq. mm, 101.3 sq. mm, and 58.9 sq. mm, respectively. These values correspond to circular areas with diameters of 4.0 mm, 3.7 mm, 11.4 mm and 8.7 mm. The symmetrical shapes of the curves with the logarithmic abscissa indicate that the distributions are positively skewed. The fact that the curves are uniform in slope but shifted along the abscissa implies that the dispersions are approximately proportional to the medians. Thus, the sizes of the RA and SA I fields are about one-fifth of the SA II fields, which in turn are about half as large as the PC fields.

There were no statistically significant differences between the sizes of the fields in the three regions of the glabrous skin, except for the RA units ($P < 0.001$,

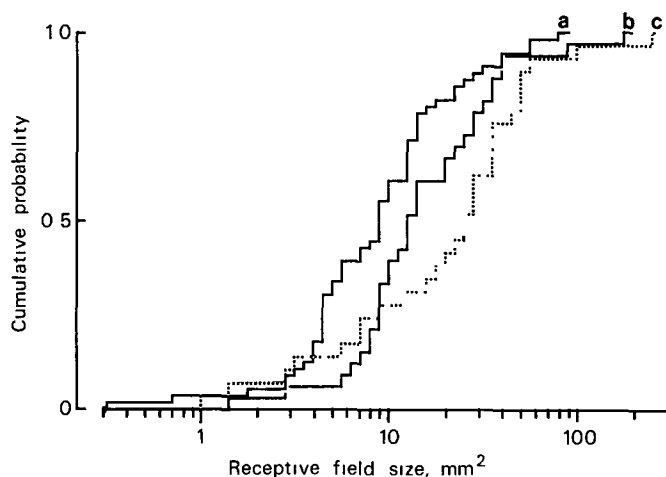


Fig. 2. Cumulative histograms of the sizes of RA receptive fields. The curves a, b and c refer to three different skin regions: the finger tip, the main part of the finger and the palm. The number of units underlying the distributions is for a 56, for b 33 and for c 29. Note that the abscissa is logarithmic.

Kruskal–Wallis). In this case there was an increase in size from the terminal phalanx to the rest of the finger and from the finger to the palm (Fig. 2). The medians were 9.4 sq. mm, 12.6 sq. mm and 28.1 sq. mm respectively. The RA fields in the finger tip were significantly different from the RA fields in the remaining part of the finger ($P < 0.05$, Mann–Whitney). On the other hand, the difference between the RA fields in the palm and the fields in the base and middle phalanges taken together was not statistically significant.

Relation between receptive field sizes and thresholds. A positive correlation was present between receptive field size and threshold, i.e. units with higher thresholds tended to have larger receptive fields. This was true for all the four unit types. Although the correlations were not all that impressive they were statistically significant ($P < 0.05$ in all cases, Spearman) except for the PC units. The RA units from the three skin regions were analysed separately in this respect (cf. Fig. 2).

Model of the population of RA and SA I units

This section is concerned with reconstructions of some population properties of the RA and SA I units. A simplified model was constructed with regard to (i) the amount of overlap of receptive fields and (ii) the layout of the receptive fields over the skin surface. On the basis of this model some aspects of the population responses to stimuli of defined features will be deduced.

The reconstructions are based on experimental data and simple assumptions. Two sets of data are involved: (i) estimates of the absolute unit densities and (ii) measurements of the field size of a sample of units as presented in the previous section. The first set of data has been presented in an earlier report where it was shown that the density of RA and SA I units vary from one skin region to the other¹⁴. The density is low in the palm (33 units/sq. cm), somewhat higher in the main part of the finger (67

units/sq. cm) and considerably higher in the finger tip (211 units/sq. cm). The values within brackets refer to densities of the two unit types taken together. These density values are based on two sets of experimental data, (i) the composition of a large sample of single units ($n = 334$) randomly collected from the median nerve and (ii) histological analysis of the total number of cutaneous nerve fibres of the kind innervating the four types of mechanosensitive units in the median nerve. Moreover, the reconstructions of the population properties are based on the simplification that the size of any receptive field is independent of stimulus intensity and equal to the size as measured with a von Frey hair at 4–5 times threshold. This is clearly an approximation but for stimuli of moderate intensity the error would not be large because the receptive fields have fairly distinct boundaries¹².

Successive steps in the development of the model required additional assumptions which will be introduced as the steps are presented.

Calculations were carried out for the three separate skin regions as defined in Methods: the finger tip, i.e. the distal half of the terminal phalanx, the main part of the finger and the palm.

Amount of total overlap of RA and SA I receptive fields. The mean overlap of fields was calculated on the basis of the unit sample collected from the appropriate skin area. The individual unit of the sample was denoted by an integer, i ($i = 1, 2, 3, \dots, n$) in order from smaller to larger fields for purposes which will be considered below. The area of the individual field was denoted a_i . The following equation gives the mean overlap (T):

$$T = \frac{D}{n} \sum_{i=1}^n a_i \quad (1)$$

where D is the estimated absolute density of the type of unit considered in the skin region¹⁴ and n is the size of the unit sample. The equation simply implies that the mean amount of overlap is equal to the absolute density of units times the mean field size of the unit sample.

Fig. 3 shows the mean overlap of RA and SA I units and of the two types taken together. The overlap in the RA populations was 20 in the fingertip and 9 in the remaining part of the finger as well as in the palm. The overlap of the SA I units in the three regions was 14, 6 and 2 respectively.

A model of the layout of the receptive fields over the skin surface. A reasonable assumption is that the small as well as the large fields of any unit type are evenly distributed over the skin surface. A slightly more extreme assumption was adopted for the purpose of producing a simplified model of the cutaneous innervation. It was assumed that the smallest fields in the population are laid out as a mosaic which completely covers the skin surface, thus constituting what will be called a sheath of fields. Moreover, sets of successively larger fields were assumed to be arranged in the same way, constituting additional sheaths. The total population of fields of one unit type would thus be arranged as a number of sheaths, the number being equal to the amount of overlap (Eqn. 1 and Fig. 3). The fields of the individual sheath would be fairly uniform in size when the amount of overlap is high.

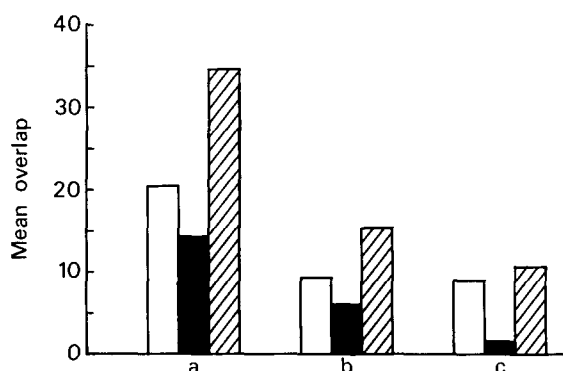


Fig. 3. Estimated mean overlap of receptive fields within three skin regions: a, finger tip; b, main part of finger; and c, palm. White and black columns refer to RA and SA I units, respectively, and hatched columns refer to the two unit types taken together.

To arrive at an estimate of the sizes of the fields constituting the separate sheaths, a first step was to calculate the amount of overlap, Q , provided by fields smaller or equal to a_i using the following equation:

$$Q = \frac{D}{n} \sum_{k=1}^i a_k \quad (2)$$

where D , a_i and n have the same meaning as in Eqn. 1 and k defines sampled field areas smaller than or equal to a_i .

Equation 2 provides a discontinuous function with one value of Q for each unit in the sample. By linear interpolation a continuous function was derived by joining the values in Q for each a_i . From this continuous function the size range of the fields which constitute the successive sheaths were estimated by calculating the field sizes corresponding to integral intervals of Q .

The estimates indicate, for instance, that RA fields smaller than 4.6 sq. mm are available in a sufficient amount to blanket the total area of the finger tip. In the palm the corresponding figure was 22.6 sq. mm.

A simplified model of the layout of the receptive fields was then constructed on the basis of the approach presented above. The individual sheath was modified on the basis of some additional assumptions. All fields constituting a sheath were assumed to be (i) circular, (ii) of uniform size and (iii) regularly spaced, border to border, with their centres in a hexagonal array, in which each centre was equally spaced from its 6 nearest neighbours. The size of these circular model fields constituting a modified sheet was set equal to the median field of the appropriate sheet as estimated from the continuous function derived from Eqn. 2. When the model was produced the sheaths were randomly positioned relative to each other in a single plane corresponding to the skin surface. The model was programmed into a desk calculator and the output was displayed by means of a plotter. Fig. 4A and B show the locations of the *centres* of the RA and SA I fields within model areas of the finger tip and the palm.

The assumptions involved may appear drastic but, in fact, the first and the

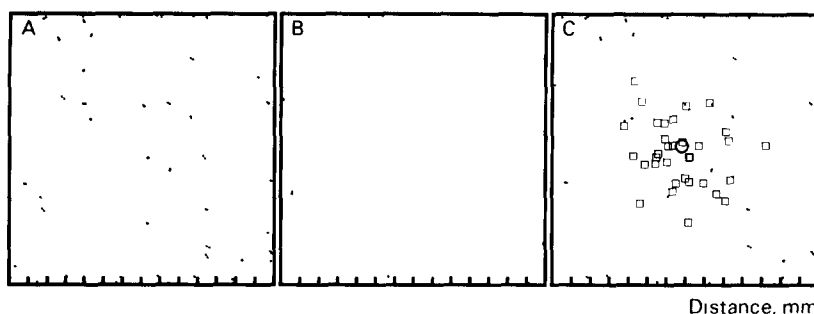


Fig. 4. Maps of model areas of the skin showing locations of RA and SA I receptive fields. A: model area of the finger tip. The *centres* of individual fields are indicated by dots, whereas the extent of the fields are not indicated. B: model area of the palm. Fields indicated as in A. C: spatial distribution of fields indented by a point stimulus with the target point in the centre of the circle. Same area as in A, with the *centre* of the fields indented indicated by squares and fields which are not indented indicated by dots.

second assumption above (i and ii) imply only a moderate deviation from experimental data for skin regions where the overlap of units is high. It might be worth pointing out that the assumptions have the consequence that the areas between the circular fields of the individual sheath were not taken into account. The sum of these areas amount to about 10 (9.31) per cent of the total sheath area.

Number and locations of fields indented by stimuli of defined geometries

On the basis of the model described in the previous section the number and the locations of fields that would be indented by stimuli of defined geometries were displayed graphically.

As a consequence of the approximation that the areas between the circular fields in the model were not taken into account the estimated number of fields indented by any stimulus is somewhat dependent on the exact target point. Therefore a range of values is given in some of the following estimates. This variation should not be taken as a description of the true variation when the skin is stimulated. Also the mean number of units activated by any stimulus would be slightly too low (about 10%).

Point stimuli. This term refers to fictitious stimulations with an infinitely small probe. The spatial distribution of fields indented by a point stimulus at a randomly chosen target point in the model of the finger tip is shown in Fig. 4C. It may be seen that the 34 fields indented are spread within a considerable area. The distance between the *centres* of the extreme fields is as large as 8 mm.

It may be argued that point indentations as defined above are of minor interest because such stimuli can not be delivered in practice. However, it can easily be shown that the number of fields indented by discs with diameters below 2 mm is not greatly influenced by the exact size of the disc. This is illustrated in Fig. 5 which shows the mean number ($n = 5$) of fields indented as a function of the disc diameter.

Pairs of point stimuli. To elucidate the capacity of spatial resolution of the system, estimates were made of the number of units indented by point indentations located closely together. Fig. 6 shows the locations of *centres* of fields indented by two

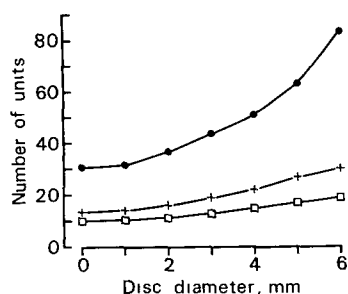


Fig. 5. Estimated mean of number of RA and SA I fields indented as a function of disc diameter ($n = 5$). The curves refer to three skin regions: finger tip (●), remaining part of the finger (+) and palm (□). The curves were drawn by joining the symbols with straight lines.

point stimuli. The two circles in the diagrams represent two target points, 1 mm, 2 mm, 4 mm and 8 mm apart as indicated. Units indented by one of the two points are indicated by squares and units indented by the other by crosses. The diagrams in Fig. 6A–C refer to the finger tip and those in D–F to the palm. It may be seen, for instance, that in the finger tip, when the distance between the target points was 1 mm, 8 and 10 units were indented exclusively by one and the other stimulus (Fig. 6A). As expected the patterns are markedly different in the palm compared to the finger tip due to the low unit density and the fact that the fields of the dominating unit type (RA) are larger (Fig. 6D–F).

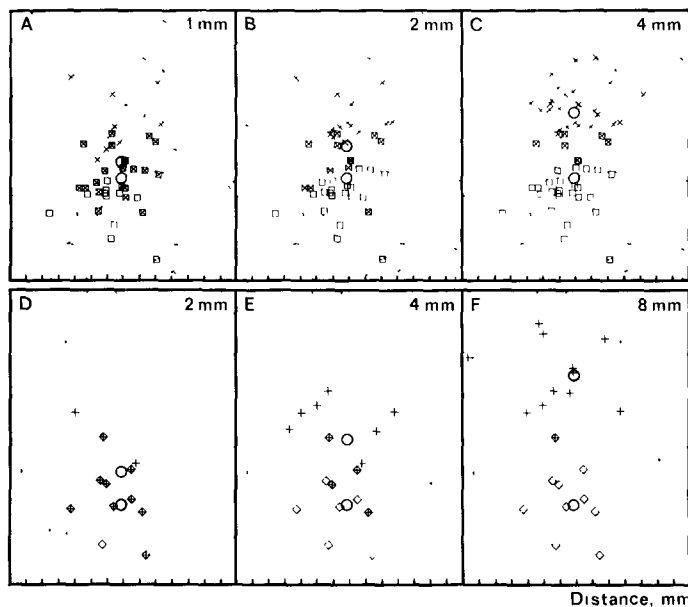


Fig. 6. Maps of model areas of the skin showing spatial distribution of RA and SA I receptive fields indented by pairs of point stimuli. Circle centres represent target points. Interpoint distance is given at the upper right of the individual map. Squares and crosses indicate *centres* of fields indented by the two stimuli respectively, and dots indicate *centres* of fields which are not indented. A–C: model areas of the finger tip. D–F: model areas of the palm.

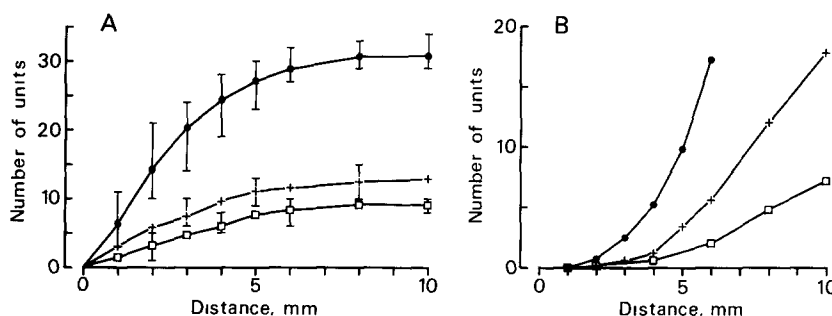


Fig. 7 A: estimated number of RA and SA I fields indented exclusively by one point stimulus of a pair as a function of the interpoint distance. Symbols indicate mean values and vertical bars the ranges ($n = 10$). B: estimated difference in mean ($n = 5$) number of RA and SA I fields indented by a bar and a pair of points as a function of bar length and interpoint distance. The ends of the bar coincide with the two target points. The curves which refer to three skin regions: finger tip (●), remaining part of the finger (+) and palm (□), were drawn by joining the symbols with straight lines

Fig. 7A illustrates the number of fields indented exclusively by one of two point stimuli as a function of the interpoint distance. Each symbol gives the mean of the values obtained at 5 randomly chosen pairs of loci and the bars give the range. The three curves represent data from the three different skin regions as detailed in the legend.

Bar indentations versus pair of point stimuli. The sets of fields indented by a bar and a pair of points were compared when the bar extended between the two target points. The width of the bar was assumed to be small in relation to the size of the fields and therefore negligible for the estimates. Fig. 7B shows the mean number ($n = 5$) of fields indented exclusively by the bar as a function of the length of the bar or interpoint distance. The three curves refer to the finger tips, the main part of the fingers and the palm. It may be seen that the distance associated with a mean difference of one unit is about 2.2 mm, 3.7 mm and 4.6 mm for the three regions.

DISCUSSION

Experimental findings

Receptive field size. The receptive fields were smaller than those reported in earlier studies of mechanoreceptor units in the glabrous skin of the human hand^{18–20}. The difference is presumably accounted for by differences in experimental technique. In the earlier studies, the field borders were determined manually with a glass rod implying that the stimuli in general were much stronger than 4–5 times the threshold force as employed in the present study. This would account for a difference in the direction found because the size of the receptive field grows with the intensity of the stimulus (e.g. refs. 3, 12, 17 and 23). However, this effect is very small for the RA and SA I fields because of their distinct field borders, and it seems unlikely that differences in stimulus intensity accounted for all the discrepancies in the field size. Higher accuracy are probably easier to achieve with the more delicate instruments used in the present study.

A comparison between data from humans and subhuman primates as reported in the literature indicates that RA and SA I fields are larger in man^{18–20,22,27}. This is also true for the present findings. The subhuman primates studied in this respect obviously have smaller hands with more narrow papillary ridges. It seems feasible that the endings of the RA and SA I units are scattered with a distance between the endings proportional to the breadth of the papillary ridges.

Earlier studies of RA and SA I units in the primate hand have shown that the fields tend to be smaller the more distally they are located^{19,20,22,27}. In the present study this tendency could be demonstrated only for RA units.

Model of the population of RA and SA I units

The present report is the third in a series on the mechanosensitive units in the glabrous skin area of the human hand^{12,14}. The experimental data presented in the three studies allow an inference of some population properties. The reconstructions were limited to the RA and SA I units for two reasons. It seems likely that these units account for the pronounced spatial acuity of the tactile sense in the human hand^{12,14,29}. Moreover, the available amount of experimental data was larger for the RA and SA I units than for the PC and SA II units.

The model was used to estimate the number of fields indented by stimuli of defined geometries and relative locations of the indented fields, whereas the amount of activity in the individual unit was not considered. It deserves emphasis that the estimates should be regarded as first approximations and attention should be paid to the order of magnitude of the values rather than to the exact figure.

The model is based on the assumption that the skin region considered is uniform with regard to unit density and size distribution of receptive fields. This assumption seems reasonable as there is nothing to suggest a pronounced spatial segregation in these respects. Likewise the model is based on the approximation that the fields are circular and that their size is independent of stimulus intensity.

The concept of central field. Several considerations seem to justify the simplification that any RA or SA I field is of constant size. It has been shown in an earlier report that the RA and SA I fields have very distinct borders¹². The sensitivity decreases drastically with the distance from the field border as assessed with von Frey stimuli at 5 times threshold. For instances, at a distance of 0.5 mm from the field border, the sensitivity may be but a tenth of the maximal sensitivity of the unit. It seems therefore a reasonable approximation to regard the size of the fields as independent of stimulus intensity within limits. Weak stimuli are usually employed in tactile discriminative tests which makes the approximation particularly valid here.

As pointed out in Methods the field as defined in the present study, i.e. with von Frey stimuli at 4–5 times threshold, coincides approximately with the area where the terminals of the unit probably are located¹². This area will be denoted the central field of the unit. The total neural input to the central nervous system in response to any indentation will be dominated by activity in units whose central fields are directly indented, because the sensitivity decreases steeply with distance outside the central field. As a consequence the intensity of the response decreases as well very steep-

ly^{15,16,24}. It seems of interest in this context to speculate on the possible role of lateral inhibition in the central nervous system organized in a surround fashion (for refs., see refs. 4, 9 and 25). Since the afferent response is, on the average, more intense from the units indented within their central fields than from the other units, it follows that the average inhibitory effect would be stronger from central field units upon surrounding ones than in the opposite direction. Moreover, the response latency of any unit is longer, the longer the distance between its central field and the point of indentation^{1,15,16,18}. Therefore, many units responding to a stimulus indenting the skin surface outside their central fields would be subjected to lateral inhibition, already when their first impulses arrive to the central terminals. These two aspects suggest that the most relevant information concerning the spatial features of a stimulus is carried in units which are excited by direct indentation of their central fields.

It may be of interest to point out that the intensity of stimuli required to excite all central fields overlapping any point is not very large compared to most stimuli which are of interest to the intact organism. For instance, it can be inferred from threshold studies that indentations with amplitudes of about one mm would excite practically all the RA and SA I units¹³. Likewise, the distributions of the force thresholds indicate that roughly 90% of the RA and SA I units are excited by a 5 mN force¹⁵.

Spatial resolution of the peripheral mechanoreceptive organization. The reconstructions of the sets of receptive fields indented by point stimuli describe properties which are of interest in relation to tactual acuity, as studied in psychophysical tests of for instance point localization and two point discrimination.

The spatial patterns of fields indented by two points (Figs. 6 and 7A) may constitute the basis of localization and spatial discrimination when two points on the skin are stimulated in succession. A distinct result of the estimates was that two separate point stimuli indented many different fields even when the interpoint distance was smaller than the average diameter of the smallest fields constituting a set which might cover the model area (cf. Figs. 1, 2 and 6). Thus it seems that the spatial resolution capacity of the peripheral organization is considerably higher than indicated by the size of the small fields in the population.

It seems of interest to compare the reconstructions presented in Results with the findings in psychophysical studies on spatial acuity. An important test of the model is whether it provides the minimal information required for the psychophysical capacity.

As to the capacity of point localization, the mean error in psychophysical tests is large in relation to the capacity of the peripheral organization as predicted by the model (e.g. ref. 32). Hence the RA and SA I systems seem to provide information in excess of what is required for the psychophysical capacity in this test. It may be relevant that this type of psychophysical test engages memory to an extent that is not the case in, for instance, two point threshold tests. This might account for a poor performance in relation to the capacity of the peripheral organization. The mean error of localization, as determined with psychophysical methods, is actually in the same order of magnitude as the average diameter of the smallest fields constituting a set which might cover the skin surface in question (cf. ref. 32 and Figs. 1 and 2).

The two point threshold as defined in psychophysical tests, on the other hand, is of the same order of magnitude as the capacity of the model to discriminate between a bar and a pair of point stimuli (e.g. refs. 21 and 32). However, the standard psychophysical test procedure does not involve this type of discrimination. The subjects are usually asked to report whether they clearly feel the stimulus as two points or not, when the interpoint distance is altered. It seems reasonable to assume that the sensation of two distinct points is based on a low impulse activity in units whose fields are located partly or completely inbetween the stimulus target points, as opposed to the high activity in units whose central fields are indented by the points. Such a dip in activity would require that, at least one, non-indented central field is located between the two indentations. Thus it seems that the discriminative capacity of the model is similar to the capacity of human subjects in tests of two point threshold. On the other hand, it should be noticed that the sensations reported in two point threshold tests do not necessarily fall into two distinct groups with either a sensation of one single point or of two separate points. There are a variety of impressions ranging from a small area to a line (e.g. ref. 7). It seems that these findings are consistent with the present model which indicates that the number and the locations of the fields indented by a pair of points change considerably with interpoint distance in the range shorter than the distance required to accommodate a whole central field between the two point stimuli.

The psychophysical finding that the two point threshold is smaller in the transverse direction of the hand than in the longitudinal direction^{10,21,30} is consistent with the finding that many oval fields are oriented longitudinally. Interestingly this was already predicted by Weber in 1852³¹.

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